DATA STRUCTURES AND ALGORITHMS

U25ADG02 • Units 1–5 • Part A – 1 Mark Multiple-Choice Questions with Answers

KPR Institute of Engineering and Technology | Semester III | R2025

UNIT I – LISTS, STACKS AND QUEUES

1. A stack checks balanced parentheses in "{[()]}". What is the maximum stack size reached?

a) 2 **b)** 3 **c)** 4 **d)** 1

Correct answer: c) 4

2. Convert infix A+B*C to postfix.

a) ABC*+ **b)** AB+C* **c)** A+BC* **d)** ABC+*

Correct answer: a) ABC*+

3. Evaluate postfix expression 6 3 2 * + 5 -.

a) 5 **b)** 6 **c)** 7 **d)** 8

Correct answer: c) 7

4. Circular queue size 6: front=1, rear=3 (3 elements). Two enqueues then one dequeue. New rear?

a) 3 **b)** 4 **c)** 5 **d)** 0

Correct answer: c) 5

5. Insert 25 after 20 in 10→20→30→40. What is at position 3 (1-indexed)?

a) 20 **b)** 25 **c)** 30 **d)** 40

Correct answer: b) 25

6. Delete 30 from circular list 10→20→30→40→back to 10. Which value follows 20?

a) 10 **b)** 20 **c)** 30 **d)** 40

Correct answer: d) 40

7. In A↔B↔C↔D, delete C. What does B.next point to?

a) A **b)** C **c)** D **d)** NULL

Correct answer: c) D

8. Insert 25 at index 2 (0-indexed) in [10,20,30,40,50]. Result?

a) [10,20,25,30,40,50] **b)** [10,25,20,30,40,50] **c)** [10,20,30,25,40,50] **d)** [25,10,20,30,40,50]

Correct answer: a) [10,20,25,30,40,50]

9. Add $3x^2+5x+2$ and $2x^2+4$ using Polynomial ADT.

a) $5x^2+5x+6$ **b)** $5x^2+9x+2$ **c)** $6x^2+5x+6$ **d)** $5x^2+5x+2$

Correct answer: a) $5x^2+5x+6$

10. In an LSD radix-sort units pass, which bucket receives 23?

a) Bucket 2 **b)** Bucket 3 **c)** Bucket 4 **d)** Bucket 5

Correct answer: b) Bucket 3

11. Evaluate prefix expression * + 2 3 4.

a) 9 **b)** 14 **c)** 20 **d)** 24

Correct answer: a) 9

12. Convert (A+B)*(C-D) to postfix.

a) AB+CD-* **b)** AB+CD*- **c)** A+B*C-D **d)** ABCD+-*

Correct answer: a) AB+CD-*

13. Push then pop every character of STACK. What is the output?

a) STACK **b)** KCATS **c)** CATSK **d)** TACKS

Correct answer: b) KCATS

14. Deque [10,20,30]: insert 5 at front and 40 at rear. Result front-to-rear?

a) 5,10,20,30,40 **b)** 10,20,30,5,40 **c)** 40,10,20,30,5 **d)** 5,40,10,20,30

Correct answer: a) 5,10,20,30,40

15. Enqueue A,B,C, then dequeue twice. What is the front element?

a) A **b)** B **c)** C **d)** Empty

Correct answer: c) C

16. Sparse-matrix row 2 has nonzeros at columns 1 and 4. Number of data nodes in that row?

a) 1 **b)** 2 **c)** 3 **d)** 4

Correct answer: b) 2

17. Using stack symbol balancing, what is the result for "(a+b]"?

a) Balanced **b)** Unbalanced **c)** Depends on operator **d)** Cannot be determined

Correct answer: b) Unbalanced

18. Array List ADT capacity 5 is full. What is needed before inserting a sixth element?

a) Delete first element **b)** Resize/reallocate array **c)** Insert directly **d)** Convert to linked list

Correct answer: b) Resize/reallocate array

19. Circular queue size 4: front=2,rear=2, one element. After one dequeue, state?

a) Full **b)** Empty **c)** Overflow **d)** Unchanged

Correct answer: b) Empty

20. Convert A*(B+C)-D to postfix.

a) ABC+*D- **b)** AB+C*D- **c)** ABCD+*- **d)** A*BC+D-

Correct answer: a) ABC+*D-

UNIT II – SEARCHING, SORTING AND HASHING

1. Binary search [11,15,130,145,152,170,185]: first mid value (0-indexed)?

a) 130 **b)** 145 **c)** 152 **d)** 170

Correct answer: b) 145

2. Worst-case comparisons for linear search of an absent item in 7 elements?

a) 3 **b)** 6 **c)** 7 **d)** 1

Correct answer: c) 7

3. After one Bubble Sort pass on [5,3,8,1], what is the array?

a) [3,5,1,8] **b)** [3,5,8,1] **c)** [1,3,5,8] **d)** [5,3,8,1]

Correct answer: a) [3,5,1,8]

4. Insertion sort [9,4,6,2]: array after processing index 1?

a) [4,9,6,2] **b)** [9,4,6,2] **c)** [2,4,6,9] **d)** [4,6,9,2]

Correct answer: a) [4,9,6,2]

5. Selection sort [29,10,14,37]: which element is placed first after pass 1?

a) 29 **b)** 10 **c)** 14 **d)** 37

Correct answer: b) 10

6. Quick sort, first pivot, [50,30,90,10]: pivot position after partition?

a) Index 0 **b)** Index 1 **c)** Index 2 **d)** Index 3

Correct answer: c) Index 2

7. Merge sort [8,4]: what is the first step?

a) Compare and swap **b)** Split into [8] and [4] **c)** Insert 4 before 8 **d)** Select minimum

Correct answer: b) Split into [8] and [4]

8. Hash table size 10, h(k)=k mod 10, separate chaining: key 25 goes to?

a) Bucket 2 **b)** Bucket 5 **c)** Bucket 0 **d)** Bucket 15

Correct answer: b) Bucket 5

9. Linear probing, h(k)=3 and bucket 3 occupied: where is the new key placed?

a) Bucket 3 **b)** Bucket 4 **c)** Bucket 13 **d)** Rejected

Correct answer: b) Bucket 4

10. Load factor exceeds threshold. Which operation restores performance?

a) Chaining **b)** Rehashing **c)** Linear search **d)** Sorting

Correct answer: b) Rehashing

11. Binary search on 6 elements: maximum iterations to find any element?

a) 2 **b)** 3 **c)** 6 **d)** 1

Correct answer: b) 3

12. Selection sort [4,2,7,1]: after how many passes is it guaranteed sorted?

a) 1 **b)** 2 **c)** 3 **d)** 4

Correct answer: c) 3

13. Quick sort first pivot on [4,1,3,9,7]: elements in left partition after first partition?

a) 1,3 **b)** 1,3,9 **c)** 9,7 **d)** None

Correct answer: a) 1,3

14. Insertion sort [1,2,4,3,5]: shifts needed to place 3 correctly?

a) 0 **b)** 1 **c)** 2 **d)** 3

Correct answer: b) 1

15. Extendible hashing: bucket overflows and local depth equals global depth. What happens?

a) Discard key **b)** Split bucket and double directory **c)** Delete bucket **d)** Ignore overflow

Correct answer: b) Split bucket and double directory

16. Hash size 7, h(k)=k mod 7: which of 10,17,24 collide?

a) 10 and 17 **b)** 17 and 24 **c)** 10,17 and 24 **d)** None

Correct answer: c) 10,17 and 24

17. Merge sorted lists [2,5,8] and [1,6,9]: first element in merged result?

a) 2 **b)** 5 **c)** 1 **d)** 8

Correct answer: c) 1

18. Bubble sort on already sorted 5-element array with early exit: passes executed?

a) 1 **b)** 4 **c)** 5 **d)** 0

Correct answer: a) 1

19. Quadratic probing, h(k)=k mod 10, collision at slot 4: next probe using $+1^2$?

a) Slot 5 **b)** Slot 6 **c)** Slot 8 **d)** Slot 0

Correct answer: a) Slot 5

20. Binary search for 100 in [10,20,30,40,50]: outcome?

a) Found at index 4 **b)** Found at index 0 **c)** Not found **d)** Infinite loop

Correct answer: c) Not found

UNIT III – TREES, BST AND AVL TREES

1. Inorder traversal of BST [50,30,70,20,40,60,80]: first value?

a) 50 **b)** 30 **c)** 20 **d)** 80

Correct answer: c) 20

2. Preorder traversal of the same BST: first value?

a) 20 **b)** 30 **c)** 50 **d)** 70

Correct answer: c) 50

3. Postorder traversal of the same BST: last value?

a) 50 **b)** 80 **c)** 20 **d)** 40

Correct answer: a) 50

4. Insert 45 into BST [50,30,70,20,40,60,80]. It attaches as child of?

a) 30 (right) **b)** 40 (right) **c)** 50 (left) **d)** 70 (left)

Correct answer: b) 40 (right)

5. Delete leaf 20 from the BST. Which node's child pointer becomes NULL?

a) 50 **b)** 30 **c)** 70 **d)** 40

Correct answer: b) 30

6. Delete node 30 with children 20 and 40. Which inorder successor replaces it?

a) 20 **b)** 40 **c)** 50 **d)** 60

Correct answer: b) 40

7. Expression tree for A+B*C: what is stored at the root?

a) A **b)** B **c)** + **d)** *

Correct answer: c) +

8. Postorder traversal of expression tree A+B*C gives?

a) ABC*+ **b)** AB+C* **c)** A+BC* **d)** ABC+*

Correct answer: a) ABC*+

9. AVL node has left height 2 and right height 0. Balance factor?

a) 0 **b)** 1 **c)** 2 **d)** -2

Correct answer: c) 2

10. Insert 10,20,30 into empty AVL tree. Which rotation restores balance?

a) LL **b)** RR **c)** LR **d)** RL

Correct answer: b) RR

11. Insert 30,10,20 into empty AVL tree. Which rotation restores balance?

a) LL **b)** RR **c)** LR **d)** RL

Correct answer: c) LR

12. Level-order traversal of root 1 with children 2 and 3?

a) 1,2,3 **b)** 2,1,3 **c)** 1,3,2 **d)** 3,2,1

Correct answer: a) 1,2,3

13. Complete binary tree with 15 nodes: height when root is level 0?

a) 2 **b)** 3 **c)** 4 **d)** 15

Correct answer: b) 3

14. Search for 60 in BST [50,30,70,20,40,60,80]: comparisons?

a) 1 **b)** 2 **c)** 3 **d)** 4

Correct answer: c) 3

15. Left height 1 and right height 3: is AVL node balanced?

a) Yes, balanced **b)** No, right-heavy imbalance **c)** No, left-heavy imbalance **d)** Cannot be determined

Correct answer: b) No, right-heavy imbalance

16. Expression tree for (A-B)/C: root?

a) - **b)** / **c)** A **d)** C

Correct answer: b) /

17. Inorder traversal of a BST produces which sequence?

a) Random order **b)** Descending order **c)** Ascending sorted order **d)** Level order

Correct answer: c) Ascending sorted order

18. Delete root 50 from BST [50,30,70,20,40,60,80] using standard successor method. New root?

a) 40 **b)** 60 **c)** 70 **d)** 80

Correct answer: b) 60

19. An LR double rotation internally performs how many single rotations?

a) 1 **b)** 2 **c)** 3 **d)** 0

Correct answer: b) 2

20. Preorder A,B,D,E,C and inorder D,B,E,A,C: root?

a) B **b)** D **c)** A **d)** C

Correct answer: c) A

UNIT IV – GRAPHS AND GRAPH ALGORITHMS

1. BFS from A on edges A-B, A-C, B-D, C-D: traversal order?

a) A,B,C,D **b)** A,C,B,D **c)** A,B,D,C **d)** A,D,B,C

Correct answer: a) A,B,C,D

2. DFS from A on edges A-B, B-C, A-D: one valid traversal (visit B before D)?

a) A,B,C,D **b)** A,D,B,C **c)** A,C,B,D **d)** D,A,B,C

Correct answer: a) A,B,C,D

3. Undirected graph edges (1,2),(1,3),(2,3): adjacency-matrix entry [1][3]?

a) 0 **b)** 1 **c)** 2 **d)** -1

Correct answer: b) 1

4. Dijkstra from A: edges A-B(4), A-C(2), C-B(1). Shortest distance A to B?

a) 4 **b)** 3 **c)** 2 **d)** 1

Correct answer: b) 3

5. DAG edges 1→2,1→3,2→4,3→4: valid topological order?

a) 1,2,3,4 **b)** 4,3,2,1 **c)** 2,1,3,4 **d)** 3,1,2,4

Correct answer: a) 1,2,3,4

6. Prim from A, edges A-B(2), A-C(3), B-C(1): first edge added?

a) A-B **b)** A-C **c)** B-C **d)** None

Correct answer: a) A-B

7. Kruskal processes weights [4,1,3,2] in what order?

a) 4,1,3,2 **b)** 1,2,3,4 **c)** 1,3,2,4 **d)** 4,3,2,1

Correct answer: b) 1,2,3,4

8. Undirected graph with 5 vertices and 4 edges forming a spanning tree is?

a) Always cyclic **b)** Acyclic (a tree) **c)** Only if weighted **d)** Cannot be determined

Correct answer: b) Acyclic (a tree)

9. BFS uses which data structure for next vertex?

a) Stack **b)** Queue **c)** Priority queue only **d)** Array

Correct answer: b) Queue

10. DFS uses which structure (explicitly or recursively) for backtracking?

a) Queue **b)** Stack **c)** Hash table **d)** Tree

Correct answer: b) Stack

11. Dijkstra relaxation when dist[u]+w < dist[v] causes what?

a) dist[v] unchanged **b)** dist[v] updated to dist[u]+w **c)** dist[u] updated **d)** Edge removed

Correct answer: b) dist[v] updated to dist[u]+w

12. Kruskal uses which structure to detect cycles?

a) Stack **b)** Union-Find (Disjoint Set) **c)** Queue **d)** Adjacency Matrix only

Correct answer: b) Union-Find (Disjoint Set)

13. Edges A-B:4,A-C:2,B-D:5,C-B:1,C-D:8,D-E:6. Dijkstra from A: shortest distance to D?

a) 5 **b)** 6 **c)** 8 **d)** 10

Correct answer: c) 8

14. DFS topological sort adds a vertex to ordering when?

a) When first visited **b)** After all descendants finish **c)** Before visiting neighbours **d)** Never

Correct answer: b) After all descendants finish

15. Prim repeatedly selects which edge?

a) Any random edge **b)** Minimum-weight edge crossing tree/cut **c)** Maximum-weight edge **d)** Last unvisited edge

Correct answer: b) Minimum-weight edge crossing tree/cut

16. Sparse graph with 100 vertices and 150 edges: lower-memory representation?

a) Adjacency Matrix **b)** Adjacency List **c)** Incidence Matrix **d)** Both equal

Correct answer: b) Adjacency List

17. Disconnected components {A,B} and {C,D}; BFS starts at A. Visited vertices?

a) A,B,C,D **b)** A,B only **c)** C,D only **d)** A only

Correct answer: b) A,B only

18. Kruskal considers an edge that would form a cycle. What does it do?

a) Add it anyway **b)** Skip it and consider next edge **c)** Stop algorithm **d)** Remove previous edge

Correct answer: b) Skip it and consider next edge

19. In directed DFS cycle detection, a back edge points to which vertex?

a) Unvisited vertex **b)** Finished vertex **c)** Vertex in recursion stack **d)** Source vertex only

Correct answer: c) Vertex in recursion stack

20. MST of a connected graph with n vertices contains how many edges?

a) n **b)** n–1 **c)** n+1 **d)** 2n

Correct answer: b) n–1

UNIT V – ALGORITHM ANALYSIS AND COMPLEXITY

1. Which asymptotic notation describes a tight bound (upper and lower)?

a) Big-O **b)** Big-Omega **c)** Big-Theta **d)** Little-o

Correct answer: c) Big-Theta

2. Which notation expresses only the worst-case upper bound?

a) Big-O **b)** Big-Omega **c)** Big-Theta **d)** Little-omega

Correct answer: a) Big-O

3. Which grows slowest: O(log n), O(n), O(n log n), $O(n^2)$?

a) O(log n) **b)** O(n) **c)** O(n log n) **d)** $O(n^2)$

Correct answer: a) O(log n)

4. What is the first step in algorithmic problem solving?

a) Coding the solution **b)** Understanding the problem **c)** Testing the program **d)** Measuring running time

Correct answer: b) Understanding the problem

5. Which problem type rearranges items into a particular order?

a) Searching **b)** Sorting **c)** Graph **d)** Combinatorial

Correct answer: b) Sorting

6. Which mathematical tool typically describes running time of a recursive algorithm?

a) Recurrence relation **b)** Truth table **c)** Hash function **d)** Probability distribution

Correct answer: a) Recurrence relation

7. Basic operation count in a non-recursive algorithm measures what?

a) Memory used **b)** Number of executions of the most significant operation **c)** Variables declared **d)** Compilation time

Correct answer: b) Number of executions of the most significant operation

8. If f(n)=O(g(n)) and g(n)=O(h(n)), then f(n)=O(h(n)): which property?

a) Symmetry **b)** Transitivity **c)** Reflexivity **d)** Additivity

Correct answer: b) Transitivity

9. Correct growth-rate order, smallest to largest?

a) $n^2 < n \log n < \log n < n$ **b)** $\log n < n < n \log n < n^2$ **c)** $n < \log n < n^2 < n \log n$ **d)** $n^2 < n < n \log n < \log n$

Correct answer: b) $\log n < n < n \log n < n^2$

10. Why analyze space complexity?

a) Code readability **b)** Estimate memory as input size grows **c)** Count functions used **d)** Compiler compatibility

Correct answer: b) Estimate memory as input size grows

11. Which case describes input taking the least time?

a) Worst case **b)** Best case **c)** Average case **d)** Amortized case

Correct answer: b) Best case

12. Which is an important problem type covered besides sorting/searching?

a) String processing **b)** Compiler design **c)** OS scheduling **d)** Database normalization

Correct answer: a) String processing

13. What technique commonly solves divide-and-conquer recurrence relations?

a) Master theorem **b)** Newton-Raphson **c)** Simplex method **d)** Euclidean algorithm

Correct answer: a) Master theorem

14. Which notation describes a lower bound on growth rate?

a) Big-O **b)** Big-Omega **c)** Big-Theta **d)** None

Correct answer: b) Big-Omega

15. Main difference in efficiency analysis of recursive and iterative algorithms?

a) Recursive uses recurrence relations; iterative counts loop iterations **b)** Both always use same technique **c)** Recursive has no complexity **d)** Iterative cannot be analyzed

Correct answer: a) Recursive uses recurrence relations; iterative counts loop iterations

16. Which factor is generally independent of programming language?

a) Order of growth of running time **b)** Exact execution time in seconds **c)** Compiler used **d)** Machine clock speed

Correct answer: a) Order of growth of running time

17. What does O(1) running time mean?

a) Linear growth **b)** Constant, independent of input size **c)** Exponential growth **d)** Undefined

Correct answer: b) Constant, independent of input size

18. What is generally first considered when framing algorithm efficiency?

a) Programming language **b)** Input size and how running time depends on it **c)** Comment lines **d)** Variable naming

Correct answer: b) Input size and how running time depends on it

19. An algorithm whose time depends significantly on input arrangement, not just size, has what?

a) Best/worst-case distinction **b)** Constant time **c)** No loops **d)** Recursive with no base case

Correct answer: a) Best/worst-case distinction

20. Main purpose of asymptotic notation?

a) Exact running time in seconds **b)** General growth behaviour independent of hardware/implementation **c)** Compare source-code length **d)** Calculate memory addresses

Correct answer: b) General growth behaviour independent of hardware/implementation